AE 162 BASIC AGRICULTURE

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COURSE OUTLINE

COURSE OUTLINE (2, 0, 2)

Crop Production

1. **Principles and practices of arable and plantation crops**
2. **Factors affecting growth and production of crops**

COURSE OUTLINE (2, 0, 2)

Crop Production …

3. **Soil/plant/water relationships**
4. **Technical and economic aspects of agriculture systems**
5. **Agronomic/agriculture practices and principles**

COURSE OUTLINE (2, 0, 2)

Animal Production

1. **The field of animal production, feeding, breeding**
2. **Introductory poultry science**
3. **Animal production mechanisation**



INTRODUCTION

DEFINITION OF AGRICULTURE

Definition of Agriculture

Agriculture is the production of crops and the rearing of animals to satisfy human needs.

Definition of Agriculture

- **Production of crops**
- **Rearing of animals**
- **Satisfaction of human needs**


HUMAN NEEDS

Human Needs

Human needs include:

- **Food**
- **Shelter**
- **Goods**
- **Employment**
- **Clothing**
- **Health**
- **Money**
- **Work**

FOOD

Food

This includes:

- **Maize**
- **Onions**
- **Cassava**
- **Plantains**
- **Pepper**
- **Sweet potatoes**
- **Yams**
- **Garden eggs**
- **Tomatoes**

Food



Food



Food …

This includes:

Cattle	**Sheep**
Goats	**Chickens**
Turkey	**Fish**

These foods are eaten in various forms.

Food ...

Food crops may also be classified as

- **Cereals**
- **Legumes**
- **Roots and tubers**
- **Oil-bearing crops**
- **Fruits and vegetables.**

CLOTHING

Clothing

- Clothing is obtained from cotton, skin, hides and hairs.
- Cotton is obtained from the cotton plant.
- Cotton is made into yarns, which are used to manufacture clothing.
- Textile plants produce various clothes.

Clothing



Cotton

Clothing





Clothing

Clothing





Clothing

Clothing …

Animals supply us with skin, hides and hair (wool).

Skin is obtained from small animals:

- **Antelopes**
- **Goats**
- **Sheep**
- **Hares**
- **Rabbits**

Clothing …

Hides are supplied by large animals:

- **Cattle**
- **Buffalo**
- **Elephants**
- **Horses**

Hides and skins are made into leather used to make belts, shoes and gloves.

Clothing …

- Hair is changed into wool which is then used to produce clothing including socks, pullovers and suits.
- They provide warmth and can be very useful in protecting the body from cold.

SHELTER

Shelter



Shelter

Shelter provides protection against:

- **Bad weather for humans**
- **Animals**
- **Crops**
- **Storage for farm products**
- **Resting places for humans and animals**

Shelter ...

Plant materials for shelter include:

- **Wawa**
- **Odum**
- **Mahogany**
- **Bamboo**

Grass can be used for roofing.


HEALTH

Health



Health …

Farm products can be used to make drugs:

- **Neem as antimalarial and pesticides;**
- **Pawpaw seeds for deworming, and**
- **Pawpaw leaves to soften meat**

Health …

- **Farm products can be used to make drugs:**
- **Animal fats, for example pig lard, for drugs in factories.**



GOODS, FOREIGN EXCHANGE AND INCOME

Goods, Foreign Exchange and Income





Goods

Most goods in Ghana are produced from raw materials which are obtained from agricultural products.

Foreign Exchange

When we produce the goods we need in our factories from agricultural raw materials, we save money because we do not have to import them.

Foreign Exchange

We also produce cash crops like cocoa, pineapple and banana.

Foreign Exchange …

We also produce cash crops like, pepper, yam and ginger for export to get foreign exchange.

Some of the countries we export to are Britain, USA, Germany and China.

Foreign Exchange …

The export receipts enable us to buy things that we do not produce in Ghana.

Income

- Income is the money we get within a certain time or period.
- When farmers produce their crops for the season, they sell some to get money.
- They get money to pay for hospital fees, school fees and buying clothing.

EMPLOYMENT

Employment



Employment



Employment



Employment



Employment



Employment

- **Employment is regular work which one does for a living.**

- **The regular work of farmers is agriculture.**

Employment …

- Other people working in agriculture are tractor operators, herdsmen and farmhands to earn a living.

- Sixty percent of Ghanaians are involved in agriculture-related employment.

PRODUCTS FROM RAW MATERIALS

Products from raw materials

RAW MATERIALS	PRODUCTS
Cocoa beans	Cocoa powder, chocolate, milo, cocoa butter
Rubber	Bicycle and vehicle tyres
Cotton	Wax prints
Hair	Wool, pullovers, socks, blankets

Products from raw materials …

RAW MATERIALS	PRODUCTS
Hooves	Glue and gelatin
Horns	Combs, buttons, handles for spoons and knives
Glands, blood	Drugs, as source of hormones (insulin)
Hides, skins	Leather for shoes, bags, belts, furniture

Products from raw materials …

RAW MATERIALS	PRODUCTS
Wool, hairs, bristles	Tent covers
Kenaf	Jute bags
Citrus	Jam, marmalade, lime juice

Products from raw materials …

RAW MATERIALS	PRODUCTS
Corn husks	Door mats, hats, nylon materials
Cane	Baskets, baby cots, furniture
Sugarcane	Sugar

Products from raw materials …

RAW MATERIALS	PRODUCTS
Coffee, Tea	Beverages
Animal fats	Glue and adhesives, candles, protective clothing, phonograph records, linoleum, synthetic rubbers

TYPES/IMPORTANCE OF FARM ANIMALS

Types/importance of farm animals

Several animals are raised in Ghana.

- **We want to study the types and importance of farm animals.**

Types/importance of farm animals

Ghanaian farmers rear:

Poultry	**Sheep**	**Goats**
Rabbits	**Cattle**	**Pigs**
Fish		

POULTRY

Poultry



Poultry



Poultry



Poultry

- Poultry are birds which are kept for their meat and eggs.
- The term poultry is used for all birds which have a pair of legs covered with scales.
- They have feathers covering their bodies and a pair of wings.

Poultry

- They lay eggs and sit on them to hatch their young ones.
- The types of poultry kept by Ghanaian farmers are the same as found throughout the world.

Poultry …

These are:

- **Chickens**
- **Turkeys**
- **Guinea fowls**
- **Duck**
- **Geese**

- **Chickens are the commonest of poultry raised in Ghana.**

LIVESTOCK - SHEEP

Sheep



Sheep

- **Three types of sheep are raised in Ghana.**
- **They are the Forest types, Nungua Blackhead and the Long-legged.**

Sheep …

The Forest Types

- **They are small in size and have short legs.**
- **The rams have horns but the ewes do not have any.**
- **Their colour is usually black and white, all black, brown or a mixture of the colours.**

Sheep …

The Forest Types

- They can survive any weather.
- They are common in the rain forest areas of Ghana like Ashanti and the Coastal areas.

Sheep …

The Nungua Blackhead

- They usually have a white body with a blackhead.
- They were developed from the forest type sheep by the University of Ghana's Agricultural Research Station at Nungua.

Sheep …

The Long-legged Sheep

- **The Long-legged sheep are very common in Northern Ghana.**
- **They have long legs and large ears.**
- **The rams are horned but the ewes are hornless.**

LIVESTOCK - GOATS

Goats



Goats



Goats

There are two types of goat in Ghana.

These are the:

- **West African Dwarf**
- **Sokoto Red**

Goats ...

The West African Dwarf

- They are common in Southern Ghana.
- They can stand many farm animal diseases but they do not like water.
- They have short legs but are meatier.

Goats ...

The West African Dwarf

- **Their colour may be a mixture of ash, black, brown or white or they can be of one colour.**

- **Adult goats of both sexes may or may not have horns.**

Goats ...

The Sokoto Red

- **They are common in Northern Ghana.**
- **They have longer legs.**
- **Both the ram and ewe have horns.**
- **They have short ears which are horizontal.**

LIVESTOCK - PIGS

Pigs



Pigs

The types of pigs in Ghana are the:

- **Ashanti Black**
- **Landrace**
- **Large White and**
- **Comborough.**



Pigs …

- **The Ashanti Black are the local types.**
- **They are common in the forest and coastal areas.**
- **They are small and short and also have short hairs.**

Pigs …

- **The Landrace, the Large White and the Comborough are the imported (exotic) types.**
- **They have longer hairs.**
- **They also grow faster and have huge bodies.**

Pigs …

- **Of the three exotic pigs, the Comborough is the latest to be introduced into the country.**

LIVESTOCK - CATTLE

Cattle

Three types of cattle are reared in Ghana.

- **Beef cattle**
- **Dairy cattle**
- **Dual-purpose**

Cattle



Cattle



Cattle



Cattle …

Beef Cattle

- These are raised for their meat.
- They have large nuzzles and wide heads.

Cattle …

Beef Cattle

- Beef cattle in Ghana are the N'dama and the West African Short Horn (WASH).

- They have a tight skin with short hair.

Cattle …

Beef Cattle

- They can resist the cattle disease called trypanosomiasis or sleeping sickness.

Cattle …

Dairy cattle

- **They are raised for their milk.**
- **They have large, soft udders.**
- **The hind-quarters are wide open to contain the udders.**

Cattle …

Dairy cattle

- **Their necks are long and thin.**

- **Dairy cattle in Ghana are the Sokoto Gudali and White Fulani.**

Cattle …

Dual-purpose cattle

- **The third type of cattle are the dual-purpose ones.**
- **They produce enough milk and good meat.**

Cattle …

Dual-purpose cattle

- **Examples are the**
  - **West African Short Horn and**
  - **White Fulani.**

LIVESTOCK - RABBITS

Rabbits



Rabbits



Rabbits

Two types of rabbits are kept in Ghana:

- **Local rabbits and Exotic rabbits**
- **The local types are smaller in size than the exotic types.**
- **The exotic types come from parts of Europe, France, America and New Zealand.**

Rabbits ...

Imported rabbits in Ghana include the:

- **California White**
- **New Zealand White**
- **Chinchilla**
- **Flemish Giant**

Rabbits …

- **On most farms the local and exotic ones are mixed to obtain types which can produce good meat and withstand the local climate.**

LIVESTOCK - FISH

Fish



Fish



Fish

- **Fish agriculture is becoming popular in the country because of recent campaigns by government.**
- **Many fish ponds are being constructed in almost every where in Ghana.**

Fish

The types of fish being reared are:

- **tilapia and**
- **catfish (mud fish).**

LIVESTOCK – OTHER FARM ANIMALS

Other Farm Animals

- Some animals are also kept on a small-scale by farmers.
- They include horses, donkeys, and dogs.
- Currently, some farmers are rearing ostriches.

IMPORTANCE OF FARM ANIMALS

Importance of Farm Animals

Farm animals are important since they give us

- **food,**
- **clothing,**
- **power, and**
- **manure.**

Importance of Farm Animals

They are also used for

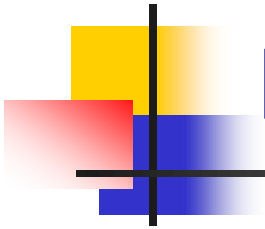
- **sports,**
- **security and**
- **medicine.**


FOOD

Food

Food products from animals reared in Ghana are:

- **meat,**
- **eggs and**
- **fish.**



Food

We obtain meat from:

- **poultry,**
- **sheep,**
- **goats,**
- **pigs.**

Food …

We obtain meat from cattle, rabbits, horses, and dogs.

We obtain milk from cattle. From milk we produce butter, cheese, yoghurt and creams.

Food ...

- In some other countries, milk from sheep and goats is also used.
- Poultry birds give eggs but chicken's eggs are the most common.
- From pigs, we get lard which is good for cooking.

CLOTHING

Clothing …

- Skin (sheep, goats, antelopes) and hides (cattle, buffalo) are used to make clothing and many types of leatherworks.
- People, especially, in Northern Ghana use leather to make various things for weaving and decorations.

Clothing …

Some of the leather goods are talismans, whisks, bracelets, amulets, mats, foot pillows, sandals, shoes, purses, bags and drums.

Clothing …

- In some temperate countries, sheep hair (wool) is used to make woolen materials.
- Sheep and goat skins are used to make carpets.
- Rabbit fur is used for making clothing in temperate countries and for making decorations.

Clothing …

- **The hairs of pigs (bristles) are used to make brushes.**


POWER

Power

- Bullocks, bulls and horses are used as sources of power.
- In the Northern parts of Ghana, some people use the donkey as a source of power.
- They may all be used to help with transportation and work on farms.

Power

- Government institutions are being encouraged to use animals for ploughing.
- The use of animals will save much foreign exchange on machinery, fuel and spare parts.
- Animals are strong and can work harder, and for longer periods than tractors.

Power …

- **We must therefore encourage their use especially in the North and the coastal areas where the land is level.**

MANURE

Manure

- **Droppings from the floor of the pen house, called litter, are often collected and added to the soil.**
- **They are called organic manure or farmyard manure.**

Manure

- **They are very helpful to vegetable plants. They contain most of the nutrients required by plants.**

Manure ...

- **Manure from animals like cattle, sheep and goats are also used for agriculture purposes.**
- **When the animals are killed, their internal organs (offal) are added to prepare chemical fertilisers.**


SPORTS

Sports

Some animals may be kept for:

- **pleasure,**
- **sports and**
- **cultural showcasing.**

Sports

The horse is kept for several uses in Ghana:

- **It may be kept for pleasure; some people like riding on horse back for enjoyment.**
- **In the North parts in Ghanait is traditional for chiefs to ride on horse back in state.**

Sports ...



Sports ...



Sports ...

The horse in Ghana is kept for several uses:

- **The horse is also used for sports and in games, such as horse racing and polo.**


SECURITY

Security

Dogs are used to keep watch over property in homes, on farms and to hunt for meat.

Cats also kill pests like mice.

Security



MEDICINE

Medicine

Parts of animals are used to manufacture medicine and medical sutures for hospitals.

Oil is obtained from fish to make cod liver oil.